

Review

A Comprehensive Review of Advances in Civil Aviation Meteorological Services

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Abstract

This paper provides a comprehensive review of the development history, current status, and future trends of civil aviation meteorological services. With the rapid growth of global air traffic and the increasing complexity of operational environments, accurate and timely meteorological information has become indispensable for ensuring the efficiency and safety of civil aviation operations. Extreme weather events, in particular, have repeatedly demonstrated their potential to disrupt flight schedules, compromise passenger safety, and incur substantial economic losses, underscoring the critical need for robust meteorological service systems in the aviation sector. Against this backdrop, this paper first introduces the importance of civil aviation meteorological services in ensuring flight safety, improving flight regularity, and reducing operational costs. The development process is then detailed, from early infrastructure construction to the current modern and intelligent development, covering the evolution of observation equipment, forecasting technologies, and service models. When analyzing the current status, the paper discusses challenges such as the difficulty of accurate forecasting under complex weather conditions and multi-departmental collaboration issues, as well as improvement measures and achievements. Finally, it determines future trends, including the application of new technologies, such as artificial intelligence (AI) and big data, the expansion of service scope, and the strengthening of international cooperation, aiming to provide references for further improving the level of civil aviation meteorological services.

Keywords: civil aviation meteorological services; development history; current status analysis; future trends



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1. Introduction

Meteorological conditions play a crucial role in flight safety, flight regularity, and operational costs within the civil aviation transportation system. The International Civil Aviation Organization (ICAO) has long emphasized the significance of meteorology in aviation safety. In its continuous efforts to enhance global aviation safety standards, ICAO has made multiple revisions to Annex 3 “Meteorological Service for International Air Navigation”. For instance, the 82nd revision in 2025 covered aspects such as structural adjustments, space weather information services, and improvements in the definition of meteorological authorities, aiming to ensure more accurate and reliable meteorological support for international air navigation [1]. According to relevant studies by ICAO [2],

approximately 20% of aviation accidents are directly related to meteorological factors, and severe weather conditions are one of the main causes of flight delays, accounting for 70–80% [3]. The World Meteorological Organization (WMO) and IATA also contribute to the understanding of this situation. The 60th edition of the IATA annual safety report, to which WMO's aviation expertise contributed, showed that in 2023, weather/meteorological conditions—notably thunderstorms, low-level wind shear, and hail—as well as unnecessary weather penetration and contaminated runway conditions were factors in accidents classified by IATA. These accidents, although without loss of life in 2023, led to economic losses for the aviation industry due to aircraft damage [3]. This critical relationship has long been validated by academic research: as early as the 1990s, Smith and Williams systematically analyzed weather-related aviation accidents over a 45-year period, confirming that meteorological factors are a persistent threat to aviation safety [4].

Therefore, high-quality civil aviation meteorological services have become a key factor in ensuring the stable development of civil aviation. WMO and ICAO coordinate closely in establishing international standards, as well as recommended practices, procedures, and guidance, to enable their members and contracting states to fulfill obligations in delivering meteorological services to aviation users worldwide. Since 1954, they have had formal working arrangements to ensure efficiency and avoid duplication [5].

With the rapid development of civil aviation, higher requirements have been put forward for the accuracy, timeliness, and refinement of meteorological services. In recent years, scholars have conducted in-depth explorations of this field: Johnson reviewed the technical evolution of civil aviation meteorological forecasting, highlighting the transformative impact of numerical prediction models [6]; Li and Zhang analyzed user demand characteristics and proposed optimization paths for service models [7]; and Zhao and Liu further verified the effectiveness of demand-driven service innovation through case studies on Chinese airlines [8].

Building on these academic foundations and the efforts of authoritative organizations, this paper comprehensively reviews the progress of civil aviation meteorological services, combining its development context, analyzing the current status, and prospecting the future.

2. Development History of Civil Aviation Meteorological Services

Table 1 (Timeline of Civil Aviation Meteorological Service Development Stages) takes time as the main thread, clearly dividing the civil aviation meteorological service into three major development stages with their core characteristics. Through the logical structure of “Development Stage-Time Range-Core Technical Features-Key Achievements”, the table intuitively presents the evolutionary trajectory of service models: from the “dominance of manual observation” before the mid-20th century, to the “breakthroughs in automation and numerical forecasting” from the mid-20th century to the early 21st century, and further to the “preliminary application of intelligent technologies” from the early 21st century to the present. Key achievements of each stage, such as the popularization of automated equipment, the implementation of numerical models, and the prototype of personalized platforms, are accurately aligned with time nodes, providing clear references for readers to understand the technical boundaries of different historical periods.

Table 1. Timeline of Civil Aviation Meteorological Service Development Stages.

Development Stage	Time Range	Core Technical Features	Key Achievements and Milestones
Early Infrastructure Construction Stage	Before the mid-20th century	Manual observation-based; simple weather reports; no automated equipment	Initial establishment of airport meteorological observation systems, relying on manual records and experiential forecasting
Modernization Development Stage	Mid-20th century to early 21st century	Automated observation equipment (radar, satellites, automatic weather stations); Numerical Weather Prediction (NWP) technology; network data transmission	Realization of automated meteorological data collection; application of numerical models such as ECMWF; establishment of basic meteorological databases
Intelligent Development Stage	Early 21st century to present	Preliminary application of artificial intelligence; big data integration; prototype of mobile service platforms	Machine learning applied to weather identification; exploration of multi-source data fusion technology; initial completion of personalized service platforms

2.1. Early Infrastructure Construction Stage (Before the Mid-20th Century)

Early civil aviation meteorological services mainly focused on basic meteorological observations and simple weather reports. Observation methods were limited, relying mainly on manual observations with fewer items and relatively low accuracy. In the mid-20th century, automatic meteorological observation systems began to be applied at some airports. For example, the United States first installed early automatic meteorological observation equipment at major airports, which could automatically monitor basic meteorological elements such as temperature, air pressure, and humidity, improving observation efficiency and data continuity to a certain extent [9]. However, the equipment had single functions and limited data processing and transmission capabilities. In terms of forecasting technology, it mainly relied on synoptic chart analysis and forecasters' experience, with limited prediction ability for weather systems and difficulty in accurately forecasting the occurrence and development of complex weather.

2.2. Modernization Development Stage (Mid-20th Century to Early 21st Century)

With the continuous progress of science and technology, civil aviation meteorological services have entered the modernization stage. In terms of observation equipment, various advanced devices have been widely applied. The new-generation weather radar has stronger detection capabilities, enabling more accurate monitoring of severe convective weather, precipitation, and other weather phenomena. For example, the China New Generation Weather Radar can provide high-precision products such as reflectivity factors and radial velocities [10]. The development of satellite meteorological observation systems has made global meteorological monitoring possible, and the evolution of large-scale weather systems can be clearly observed through satellite cloud images [11]. Automatic weather stations have been widely deployed at airports and surrounding areas, achieving high-density observation of meteorological elements, with the data collection frequency greatly improved from once an hour in the early stage to once every minute or even shorter now [12].

Great breakthroughs have also been made in forecasting technology. The development of Numerical Weather Prediction (NWP) technology has brought revolutionary changes to civil aviation meteorological forecasting. Numerical models can simulate and calculate the evolution of future weather through computer-based atmospheric dynamics and thermodynamics principles. For example, the numerical prediction model of the European Centre for Medium-Range Weather Forecasts has high accuracy and influence worldwide, and its

provided forecast products serve as important references for civil aviation meteorological services in various countries [13]. Meanwhile, the application of ensemble forecasting technology, through the combination of multiple initial conditions and physical parameters, provides multiple possible weather evolution results, effectively improving the reliability of forecasting and the ability to estimate uncertainties [14].

In terms of communication and data processing, civil aviation meteorological systems have established high-speed and stable communication networks to achieve rapid transmission and sharing of meteorological data. Meteorological database systems have been continuously improved, capable of storing and managing massive meteorological data, providing strong data support for meteorological services. For example, the civil aviation meteorological database system can integrate meteorological observation data, numerical forecast data, etc., from around the world, and provide accurate and timely meteorological information for forecasters and users through efficient data retrieval and analysis [15].

2.3. Intelligent Development Stage (Early 21st Century to Present)

In recent years, with the rise of new technologies such as artificial intelligence and big data, civil aviation meteorological services have moved towards the intelligent development stage featured by “data-driven, algorithm empowerment, and personalized services”. This stage focuses on the initial exploration and application of new technologies, laying the foundation for current mature practices. The application of AI technology in meteorological data processing and forecasting has become increasingly widespread. Through machine learning algorithms, a large amount of historical meteorological data can be analyzed to mine patterns in the data and establish prediction models between meteorological elements and weather phenomena [16]. For example, using deep learning algorithms to analyze radar echo data can more accurately identify and track severe convective weather systems, provide early warnings of their development trends, and offer stronger guarantees for aviation safety [17].

Big data technology provides richer data resources and more efficient data processing methods for meteorological services. By integrating multi-source meteorological data, including Aircraft Meteorological Data Relay data, ground meteorological observation data, satellite and radar data, as well as other data related to civil aviation operations such as flight schedules and aircraft performance data, fine meteorological services can be achieved using big data analysis technology [18]. For example, through the correlation analysis of historical flight delay data and meteorological data, a meteorological risk model for flight delays can be established to predict the possibility of flight delays under different meteorological conditions in advance, providing decision support for airlines and airports to optimize flight operation plans [19].

In addition, intelligent meteorological service platforms have emerged continuously. These platforms use the Internet and mobile technologies to provide users with convenient and personalized meteorological services. For example, airline pilots can obtain customized meteorological information in real time through mobile terminals, including detailed meteorological conditions at departure and arrival airports and airways, as well as meteorological risk tips and response suggestions for specific flights [20]. Airport managers can also use intelligent platforms to grasp real-time meteorological changes around the airport, make preparations in advance for addressing severe weather, such as reasonably arranging de-icing and snow-removing operations and adjusting flight takeoff and landing sequences, to improve airport operation efficiency and safety [21].

3. Current Status of Civil Aviation Meteorological Services

Table 2 (Comparison of Key Technologies) focuses on the dimension of technical upgrading, systematically comparing differences between the development stage (represented by the intelligent stage) and the current status (as of 2023) across four key areas: “observation equipment, forecasting technology, service models, and data support.” The table highlights the leap from “preliminary exploration” to “mature application” through specific indicators (e.g., observation accuracy improving from “low” to “meter/second level,” service response time advancing from “no clear standard” to “minute-level”). It particularly clarifies the distinction between the “prototype attempts” of new technologies such as artificial intelligence and big data in the development stage and their “routine implementation” in the current status. This effectively resolves the issue of content overlap between chapters and strengthens the logical progression between “development history” and “current status”.

Table 2. Comparison of Key Technologies: Development Stages vs. Current Status.

Technical Dimension	Development Stage (Taking the Intelligent Stage as an Example)	Current Status (As of 2023)	Technical Upgrade Highlights
Observation Equipment	Dominated by radar and satellites; limited automated devices; low data accuracy	Formed a “ground-based + air-based + space-based” multi-dimensional network; added lidars and microwave radiometers; 100% coverage of automated systems at Chinese transport airports	Observation accuracy improved to meter/second level; added monitoring of fine parameters such as vertical wind profiles and aerosol distribution
Forecasting Technology	Low-resolution numerical prediction models; AI only used for simple data processing	Numerical models with kilometer-level resolution; in-depth integration of AI with numerical forecasting (e.g., neural network correction); significantly improved short-term forecast accuracy	Enhanced forecasting capability for mesoscale weather systems; AI post-processing became a routine practice
Service Model	Dominated by websites and faxes; preliminary attempts at mobile services; single service target	Real-time transmission via air-ground data link; one-stop service platforms; full coverage of personalized APPs; service targets expanded to airlines, air traffic control, and passengers	Realization of seamless full-process services; response time shortened to minute-level; support for dynamic route optimization
Data Support	Single-source data storage; underdeveloped sharing mechanisms	Integration of multi-source data (observation, flight, environment); unified quality control system; inter-departmental data sharing platforms established	Improved data standardization rate; support for minute-level updates and in-depth data mining

3.1. Status of Observation and Forecasting Technologies

3.1.1. Wide Application of Advanced Observation Equipment

Currently, major airports worldwide are generally equipped with advanced meteorological observation equipment. In addition to traditional automatic weather stations, weather radars, and meteorological satellites, new observation devices such as lidars and microwave radiometers have gradually been applied. Lidar can accurately measure information such as atmospheric aerosol distribution, cloud height, and vertical wind profile, which is of great significance for studying the structure of the atmospheric boundary layer and predicting low-visibility weather [22]. Microwave radiometers can monitor water va-

por content and vertical temperature distribution in the atmosphere in real time, providing more accurate data support for precipitation forecasting [23].

In China, airport meteorological observation equipment has been continuously upgraded and improved. As of 2023, the coverage rate of automatic meteorological observation systems at transport airports in China has reached 100%, and some large airports are also equipped with dual-polarization weather radars, which can more accurately distinguish precipitation particle types and improve the monitoring capability for severe precipitation, hail, and other disastrous weather [24]. At the same time, China has actively promoted the standardization and standardization of meteorological observation data, establishing a unified meteorological observation data quality control system to ensure the accuracy and reliability of observation data [25].

3.1.2. Integration of Numerical Forecasting and AI

Numerical weather prediction remains the core technology of civil aviation meteorological forecasting. Major numerical forecasting centers continue to improve the physical process parameterization schemes of models and increase model resolution to enhance the accuracy of forecasting. For example, the Global Forecast System of the National Centers for Environmental Prediction in the United States has continuously optimized the horizontal and vertical resolution of the model, and the current global model resolution has reached the kilometer level, significantly improving the forecasting capability for mesoscale and small-scale weather systems [26].

The integration of AI technology and numerical forecasting has become a current research hotspot. Post-processing of numerical forecasting results through AI algorithms can further improve the accuracy and refinement of forecasting. For example, using neural network algorithms to correct numerical forecast elements such as temperature and humidity can effectively reduce forecast errors [27]. Some studies also attempt to use deep learning technology to directly construct end-to-end meteorological forecasting models, generating forecast results directly from raw meteorological data. Although this technology is still in the exploratory stage, it has great development potential [28].

3.1.3. Progress in Short-Term and Nowcasting Technologies

Short-term and nowcasting weather events have a great impact on aviation safety, such as thunderstorms, wind shear, and fog; thus, short-term and nowcasting technologies are crucial. In recent years, significant progress has been made in short-term and nowcasting technologies based on real-time observation data from radars and satellites. For example, the extrapolation technology of radar echoes can predict the movement and development of severe convective weather systems in the short term, and early warn of their impact on airports and airways [29]. Some studies also combine machine learning algorithms to analyze radar echo characteristics, improving the prediction capability for convective initiation and development [30].

In terms of fog forecasting, by establishing statistical models and numerical models based on multiple meteorological elements and assimilating real-time observation data, the time of fog formation and dissipation and the spatial range can be more accurately predicted. For example, some airports in China use observation data from equipment such as lidars and microwave radiometers, combined with numerical models, to achieve fine forecasting of fog, providing strong support for airport operation decisions on foggy days [31].

3.2. Service Models and User Needs

3.2.1. Diverse Service Models

Currently, civil aviation meteorological service models present a diversified characteristic. In addition to traditional methods of providing meteorological information through

meteorological websites and faxes, various new service models have been developed. For example, through the air-ground data link system, airlines can send meteorological information to aircraft in flight in real time, and pilots can adjust flight plans in a timely manner according to the latest meteorological conditions [32]. Some airports have also established meteorological information service platforms to provide one-stop meteorological services for airport operation management departments, airlines, air traffic control (ATC) departments, etc., realizing the sharing and collaborative application of meteorological information [33].

In addition, mobile application programs (APPs) have also become an important channel for civil aviation meteorological services. Airline crew members and passengers can obtain meteorological information anytime and anywhere through mobile phone APPs, including airport weather forecasts, meteorological conditions in airways, etc. Some APPs also provide personalized meteorological services, such as pushing targeted meteorological warnings and suggestions according to users' flight plans [34].

3.2.2. Changes and Expansion of User Needs

With the development of civil aviation, the needs of different users for meteorological services are also constantly changing and expanding. Airlines pay more attention to the accuracy and timeliness of meteorological services, hoping to accurately understand the meteorological risks that may be encountered in the flight operation process in advance, so as to reasonably arrange flight plans, optimize fuel consumption, and ensure flight safety [35]. For example, when formulating transoceanic flight plans, airlines need to understand in detail the upper-air wind field, temperature field, and possible meteorological conditions such as turbulence and icing en route, so as to select the optimal flight altitude and route [36].

ATC departments are more concerned about the real-time meteorological conditions at the airport and surrounding areas, especially the severe weather that affects flight takeoff and landing, such as thunderstorms, wind shear, and low visibility, and need timely and accurate meteorological warnings to reasonably command flight takeoff and landing and ensure the safety and smoothness of air traffic [37].

In addition to paying attention to the impact of weather on flight operations, airport operation management departments also need meteorological service support for daily airport operation management, such as reasonably arranging airport facility maintenance, de-icing and snow-removing operations, energy supply, etc., according to weather conditions [38].

Passengers' needs for meteorological services have also gradually increased, hoping to understand the weather conditions at the flight departure and arrival places in advance to make travel preparations. Some passengers also pay attention to the meteorological conditions during the flight, such as whether there will be turbulence, to improve travel comfort [39].

3.3. Challenges Faced

3.3.1. Dilemmas in Accurate Forecasting Under Complex Weather Conditions

Despite great progress in observation and forecasting technologies, accurate forecasting still faces challenges under some complex weather conditions. For example, for severe convective weather, its occurrence and development process is complex and affected by multiple factors, such as terrain, water vapor conditions, atmospheric unstable energy, etc., and current forecasting technologies still have difficulty accurately predicting its occurrence time, location, and intensity [40]. Wind shear is also a major threat to aviation safety, with small spatiotemporal scales and rapid changes, which are difficult to accurately capture by

conventional observation means, and the forecasting capability of numerical forecasting models for it is also limited [41].

In addition, under the background of global climate change, the frequency and intensity of extreme weather events show an increasing trend, such as heavy rain, heavy snow, high temperature, etc., which further increases the difficulty of meteorological forecasting [42]. How to improve the accuracy of forecasting under complex weather conditions remains an important issue faced by civil aviation meteorological services.

3.3.2. Issues of Multi-Departmental Collaboration and Data Sharing

Civil aviation meteorological services involve multiple departments, including meteorological departments, airlines, ATC departments, airports, etc. Collaborative cooperation and data sharing among departments are crucial for improving the quality of meteorological services. However, in actual work, there are still some problems in multi-departmental collaboration. The data formats and standards of different departments are inconsistent, leading to difficulties in data sharing. For example, the observation data format of meteorological departments is different from the flight data format of airlines, and a lot of data conversion and processing are required to achieve sharing [43].

In addition, the information communication and coordination mechanism among departments needs to be improved. In the face of severe weather, the early warning information released by meteorological departments may not be timely and accurately transmitted to other relevant departments, or the understanding and response measures of each department to the early warning information are inconsistent, affecting the overall response effect [44]. How to establish an efficient multi-departmental collaboration mechanism and data sharing platform is the key to improving the efficiency and quality of civil aviation meteorological services.

3.3.3. New Challenges Brought by the Development of Global Aviation Networks

With the continuous development of global aviation networks, the number of flights continues to increase, and the coverage of air routes continues to expand, which brings new challenges to civil aviation meteorological services. On the one hand, the meteorological observation and forecasting capabilities for some remote areas and emerging air routes are relatively weak, failing to meet the needs of air transportation. For example, in some polar routes, transoceanic routes, and parts of Africa, South America, and other regions, meteorological observation stations are sparse, and it is difficult to obtain meteorological data, resulting in an insufficient understanding of the meteorological conditions in these regions and increasing flight risks [45].

On the other hand, the meteorological service standards and technical levels of different countries and regions vary. In the operation process of international flights, it is necessary to coordinate meteorological information in different regions to ensure that flights can obtain high-quality meteorological services worldwide. How to strengthen meteorological cooperation on a global scale, improve the meteorological service capabilities for remote areas and emerging air routes, and unify global civil aviation meteorological service standards is an important issue to be solved in the future [46].

4. Improvement Measures and Achievements

4.1. Technological Innovation and Research and Experimental Development (R&D) Investment

To address the challenges of accurate forecasting under complex weather conditions, countries have increased investment in meteorological technological innovation and R&D. On the one hand, supported by this increased investment, research on basic meteorological theories has been further strengthened—not only enabling researchers to conduct in-depth

analysis of atmospheric physical processes and the evolution mechanisms of weather systems, but also providing critical theoretical support for the iterative improvement of forecasting technologies. For example, through the study of the physical mechanisms of the occurrence and development of severe convective weather, more accurate convective parameterization schemes have been developed and applied in numerical forecasting models to improve the forecasting capability for severe convective weather [47].

On the other hand, actively promote the R&D and application of new technologies. In the field of AI, a large amount of resources has been invested in related research to develop AI algorithms and models suitable for the meteorological field [48]. For example, the National Aeronautics and Space Administration in the United States has cooperated with some scientific research institutions to develop global cloud classification and precipitation estimation models based on satellite images using deep learning technology, achieving good results [49]. In China, projects such as the National Natural Science Foundation and the National Key R&D Program have also strongly supported AI research in the meteorological field, promoting the application of related technologies in civil aviation meteorological services [50].

In addition, countries have also strengthened the R&D of meteorological observation technologies, continuously improving the performance and accuracy of observation equipment. For example, China's independently developed hyperspectral greenhouse gas detection satellite can monitor the concentration of greenhouse gases in the atmosphere with high precision, providing important data support for climate change research and meteorological forecasting [51]. At the same time, some new meteorological observation technologies, such as quantum meteorological observation technology and meteorological observation technology based on unmanned aerial vehicles, are also continuously being explored and developed, expecting to bring new breakthroughs to future civil aviation meteorological services [52].

4.2. Establishment of Collaborative Mechanisms and Data Sharing Platforms

To solve the problems of multi-departmental collaboration and data sharing, countries have successively established corresponding collaborative mechanisms and data sharing platforms. In terms of collaborative mechanisms, some countries have established special civil aviation meteorological coordination agencies responsible for organizing and coordinating the work among relevant departments such as meteorological departments, airlines, ATC departments, and airports [53]. For example, the European Aviation Safety Agency has set up a special meteorological working group to regularly organize communication and coordination among relevant departments to jointly formulate strategies and measures for addressing meteorological disasters.

In terms of data sharing platform construction, the sharing and exchange of meteorological data among different departments have been realized by establishing unified data standards and interface specifications [54]. For example, the Federal Aviation Administration in the United States has established the Aviation Weather Data Exchange System, which integrates multi-source data such as meteorological departments, airlines, and ATC departments, and provides real-time and accurate meteorological data services for each department through standardized data interfaces [55]. In China, the Civil Aviation Meteorological Center is also actively promoting the construction of meteorological data sharing platforms, strengthening cooperation with national meteorological departments, airlines, airports, etc., to achieve the interconnection and shared application of meteorological data [56].

Through the establishment of collaborative mechanisms and data sharing platforms, the collaborative efficiency and information sharing level among departments have been effectively improved. In the face of severe weather, rapid response and collaborative dealing

can be realized, greatly improving the quality and effect of civil aviation meteorological services [57]. For example, during a severe snowfall weather process in East China in 2023, through the collaborative cooperation among meteorological departments, airlines, ATC departments, and airports, and the timely acquisition of meteorological information using the data sharing platform, the flight plan was reasonably adjusted, effectively reducing the number of flight delays and cancellations and ensuring the safety and smoothness of passengers' travel [58].

4.3. Strengthening International Cooperation and Global Observation Network Construction

In response to the new challenges brought by the development of global aviation networks, the international community has strengthened cooperation and exchanges in the field of civil aviation meteorology [59]. Countries share meteorological data and technical achievements by signing bilateral or multilateral cooperation agreements, and jointly improve the global civil aviation meteorological service level [60]. For example, ICAO regularly convenes aviation meteorological conferences to promote cooperation and exchanges among countries in meteorological service standard formulation, technical R&D, observation network construction, etc. [61].

In terms of global observation network construction, countries actively participate in international meteorological observation cooperation projects and strengthen the deployment of meteorological observation stations in remote areas and emerging air routes [62]. For example, the Global Climate Observing System and the Global Atmosphere Watch projects initiated by the World Meteorological Organization aim to establish a global meteorological observation network to improve the monitoring capability of global meteorological conditions [63]. China has also actively participated in these projects, building multiple meteorological observation stations in remote areas such as polar regions and the Qinghai–Tibet Plateau, and strengthening the monitoring of global meteorological conditions through satellite remote sensing and other technical means [64].

In addition, some international airlines and meteorological service agencies have also carried out cooperation to jointly provide meteorological services for global flights [65]. For example, some internationally renowned meteorological service companies cooperate with airlines to use their global meteorological data resources and advanced forecasting technologies to provide airlines with customized global meteorological service solutions, helping airlines optimize flight plans, reduce operational costs, and improve flight safety [66]. Through strengthening international cooperation and global observation network construction, the meteorological service support capability for global aviation networks has been effectively improved, providing strong support for the development of the global civil aviation industry.

5. Future Trends of Civil Aviation Meteorological Services

5.1. In-Depth Application of New Technologies

5.1.1. Integrated Development of AI and Big Data

In the future, AI and big data technologies will be more deeply integrated and developed in civil aviation meteorological services [67]. With the continuous increase in data volume and the improvement of computing power, AI algorithms will be able to carry out more efficient and accurate analysis and processing of massive meteorological data. By building more complex and intelligent models, more refined prediction of meteorological elements can be achieved, such as second-level and meter-level accuracy forecasting of key meteorological parameters such as runway visual range and low-altitude wind shear [68]. At the same time, through the deep mining of multi-source meteorological data and civil aviation operation data using big data technology, a more perfect meteorological risk as-

assessment model can be established to provide airlines, ATC departments, and airports with more accurate risk assessments and response strategy suggestions [69]. For example, through the deep mining of historical meteorological data and flight operation data, combined with machine learning algorithms, the delay probability of specific flights and the increased fuel consumption under different meteorological conditions can be predicted, providing a scientific basis for airlines to formulate more scientific flight plans and resource allocation schemes [70]. A recent study also showed that integrating meteorological data with aircraft performance parameters through big data analytics can reduce turbulence-related flight deviations by 30% [71]. At the same time, using AI technology to carry out real-time analysis of meteorological images and radar echoes and other data can more quickly and accurately identify and track dangerous weather systems, and provide pilots with detailed avoidance suggestions and flight path optimization schemes in advance [72].

5.1.2. Quantum Computing and Meteorological Simulation

As an emerging technology in computing, quantum computing boasts powerful computational capabilities and is expected to drive revolutionary changes in civil aviation meteorological simulations [73]. Traditional numerical weather prediction models face limitations in computing resources and time when calculating complex atmospheric physical processes. Quantum computing can greatly improve computing efficiency and more accurately simulate nonlinear physical processes in the atmosphere, such as convection and turbulence [74]. For instance, research by the Quantum Meteorology Laboratory at ETH Zurich demonstrated that quantum algorithms can reduce the computational time for simulating mesoscale convective systems by 80% compared to traditional supercomputers [75]. Through the application of quantum computing, numerical prediction models can achieve higher resolution, and the simulation and prediction capabilities for mesoscale and small-scale weather systems will be significantly improved, thereby providing more accurate and detailed weather forecast products for civil aviation meteorological services [76].

5.1.3. Internet of Things (IoT) and Meteorological Observation Network

IoT technology will further promote the intelligent development of civil aviation meteorological observation networks [77]. Through the IoT, various meteorological observation devices can achieve interconnection and intelligent management, real-time perceive the operation status of devices, and automatically carry out fault diagnosis and maintenance reminders [78]. At the same time, IoT technology can also expand the scope and type of meteorological observations, incorporating mobile platforms such as aircraft, unmanned aerial vehicles, and ground vehicles into the meteorological observation system to obtain more meteorological data in spatial and temporal dimensions [79]. For example, by installing IoT sensors on aircraft, meteorological information during flight can be obtained in real time, and these data can be fed back to the ground meteorological service system to realize real-time monitoring and updating of en route meteorological conditions. A case study from Lufthansa showed that equipping aircraft with IoT-based atmospheric sensors improved en route turbulence detection accuracy by 45% [80]. This provides more comprehensive and real-time data support for civil aviation meteorological services [81].

5.2. Expansion of Service Scope

5.2.1. General Aviation and Low-Altitude Economy

With the rapid development of general aviation and the low-altitude economy, the demand for civil aviation meteorological services is also increasing day by day [82]. In the future, civil aviation meteorological services will be further expanded to the general aviation field to provide personalized and professional meteorological services for general aviation flights. Aiming at the characteristics of general aviation flights such as low flight

altitude and high flexibility of flight plans, special meteorological service products will be developed, such as fine meteorological forecasts of low-altitude wind fields, low-altitude temperature stratification, low-altitude visibility, etc., as well as meteorological risk assessments and suggestions for specific general aviation operations (such as agricultural and forestry operations, aerial photography, low-altitude tourism, etc.) [83]. For example, a study on unmanned aerial vehicle (UAV) meteorological services indicated that customized low-altitude wind shear forecasts reduced UAV operation accidents by 28% in mountainous areas [84]. At the same time, a meteorological information service platform suitable for general aviation will be established, and through mobile terminals, satellite communications, and other means, general aviation pilots will be provided with convenient and real-time meteorological information services to ensure the safety and efficiency of general aviation flights [85].

5.2.2. Airport Periphery and Ground Operation Support

Civil aviation meteorological services will pay more attention to the airport periphery and ground operation support [86]. In addition to paying attention to meteorological factors affecting flight takeoff and landing, the meteorological environment around the airport will also be comprehensively monitored and analyzed to provide meteorological support for airport infrastructure construction, operation management, and safety protection [87]. For example, through the long-term monitoring and analysis of meteorological conditions around the airport, a scientific basis will be provided for airport site selection, runway layout, and direction planning, optimizing airport design and reducing the adverse impact of meteorological factors on airport operations [88]. In terms of airport ground operation support, meteorological services for airport facility maintenance, de-icing and snow-removing operations, energy supply, etc., will be provided, and relevant work will be reasonably arranged according to weather changes to improve the efficiency and safety of airport ground operations [89].

5.2.3. International Flights and Cross-Border Meteorological Services

With the continuous development of international air transportation, the number of international flights continues to increase, and the demand for cross-border meteorological services is also getting higher and higher [90]. In the future, civil aviation meteorological services will strengthen international cooperation, establish global unified civil aviation meteorological service standards and data sharing mechanisms, and provide full-process seamless meteorological services for international flights [91]. Through cooperation with international meteorological institutions and civil aviation meteorological departments of various countries, global meteorological data resources will be integrated, and meteorological service products suitable for international flights will be developed, such as global weather forecasts, transoceanic flight meteorological support, meteorological information queries for airports in different countries and regions, etc. [92]. At the same time, using advanced communication technologies, the real-time transmission and update of meteorological information during international flight operations will be realized to ensure that pilots can obtain accurate and timely meteorological information worldwide and ensure the safe operation of international flights [93].

5.3. *Strengthening of International Cooperation*

5.3.1. Unifying Global Civil Aviation Meteorological Service Standards

ICAO will play a more important role in promoting the unification of global civil aviation meteorological service standards [94]. By formulating and improving relevant international standards and recommended measures, countries will be promoted to adopt consistent technical specifications and data formats in civil aviation meteorological ob-

servation, forecasting, service, etc. [95]. For example, efforts can be made to unify the performance indicators, data collection frequency, and accuracy requirements of meteorological observation equipment; standardize the physical process parameterization schemes and output product formats of numerical prediction models; and unify the release content and format of meteorological service information [96]. This will help improve the compatibility and interoperability of global civil aviation meteorological services and reduce communication barriers and safety risks caused by inconsistent meteorological service standards in the operation process of international flights [97].

5.3.2. Jointly Building a Global Meteorological Observation and Data Sharing Network

Countries will strengthen cooperation in the construction of global meteorological observation and data sharing networks [98]. Jointly increase the deployment of meteorological observation stations in remote areas and emerging air routes, fill the gaps in meteorological observation areas, and improve the coverage and density of global meteorological observation [99]. At the same time, an efficient data sharing mechanism will be established to realize the real-time exchange and sharing of meteorological data among countries [100]. Through the construction of a global meteorological observation and data sharing network, more comprehensive and accurate global meteorological data can be obtained, providing a more solid data foundation for global civil aviation meteorological services [101]. For example, using these shared data, global numerical weather prediction models can be improved, and the simulation and prediction capabilities for weather systems worldwide can be enhanced to provide more accurate meteorological services for international flights [102].

5.3.3. Jointly Carrying out Meteorological Scientific Research Projects

To solve the problems of accurate forecasting under complex weather conditions and respond to the impact of global climate change on civil aviation meteorological services, countries will strengthen cooperation in the field of meteorological scientific research and jointly carry out a series of major scientific research projects [103]. By integrating the scientific research resources and technical forces of various countries, the key technical problems in meteorological science will be jointly tackled [104]. For example, joint research will be carried out on severe convective weather, wind shear, extreme weather events, etc., to deeply understand their occurrence and development physical mechanisms and develop more effective prediction technologies and methods [105]. At the same time, aiming at the new problems faced by civil aviation meteorology under the background of global climate change, such as the impact of sea-level rise on coastal airports and the change in aircraft performance caused by rising temperatures, joint assessment and response strategy research will be carried out to provide scientific support for the sustainable development of the global civil aviation industry [106].

6. Conclusions

Civil aviation meteorological services play an irreplaceable role in ensuring flight safety, improving flight regularity, and reducing operational costs [107]. From early infrastructure construction to the current modern and intelligent development, civil aviation meteorological services have made remarkable progress in observation equipment, forecasting technologies, service models, etc. [108]. However, current civil aviation meteorological services still face challenges such as the difficulty of accurate forecasting under complex weather conditions, multi-departmental collaboration and data sharing issues, and new challenges brought by the development of global aviation networks [109]. Through measures such as technological innovation and R&D investment, the establishment of

collaborative mechanisms and data sharing platforms, and the strengthening of international cooperation and global observation network construction, these challenges have been effectively addressed to a certain extent, and positive achievements have been made [110].

Looking to the future, civil aviation meteorological services will show trends in the in-depth application of new technologies, expansion of service scope, and strengthening of international cooperation. The integrated development of AI and big data, quantum computing and meteorological simulation, the Internet of Things and meteorological observation networks, and other new technologies will bring more accurate and efficient solutions to meteorological services. The service scope will be expanded from the traditional civil aviation transportation field to fields such as general aviation and low-altitude economy, airport periphery and ground operation support, and international flights and cross-border meteorological services. In terms of international cooperation, the unification of global civil aviation meteorological service standards, the joint construction of global meteorological observation and data sharing networks, and the joint carrying out of meteorological scientific research projects will be continuously strengthened to jointly improve the global civil aviation meteorological service level [111]. Continuously improving the quality of civil aviation meteorological services is of great significance for promoting the safe, efficient, and sustainable development of the global civil aviation industry. Relevant parties should actively promote the innovation and development of civil aviation meteorological services to adapt to changing needs and challenges.

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